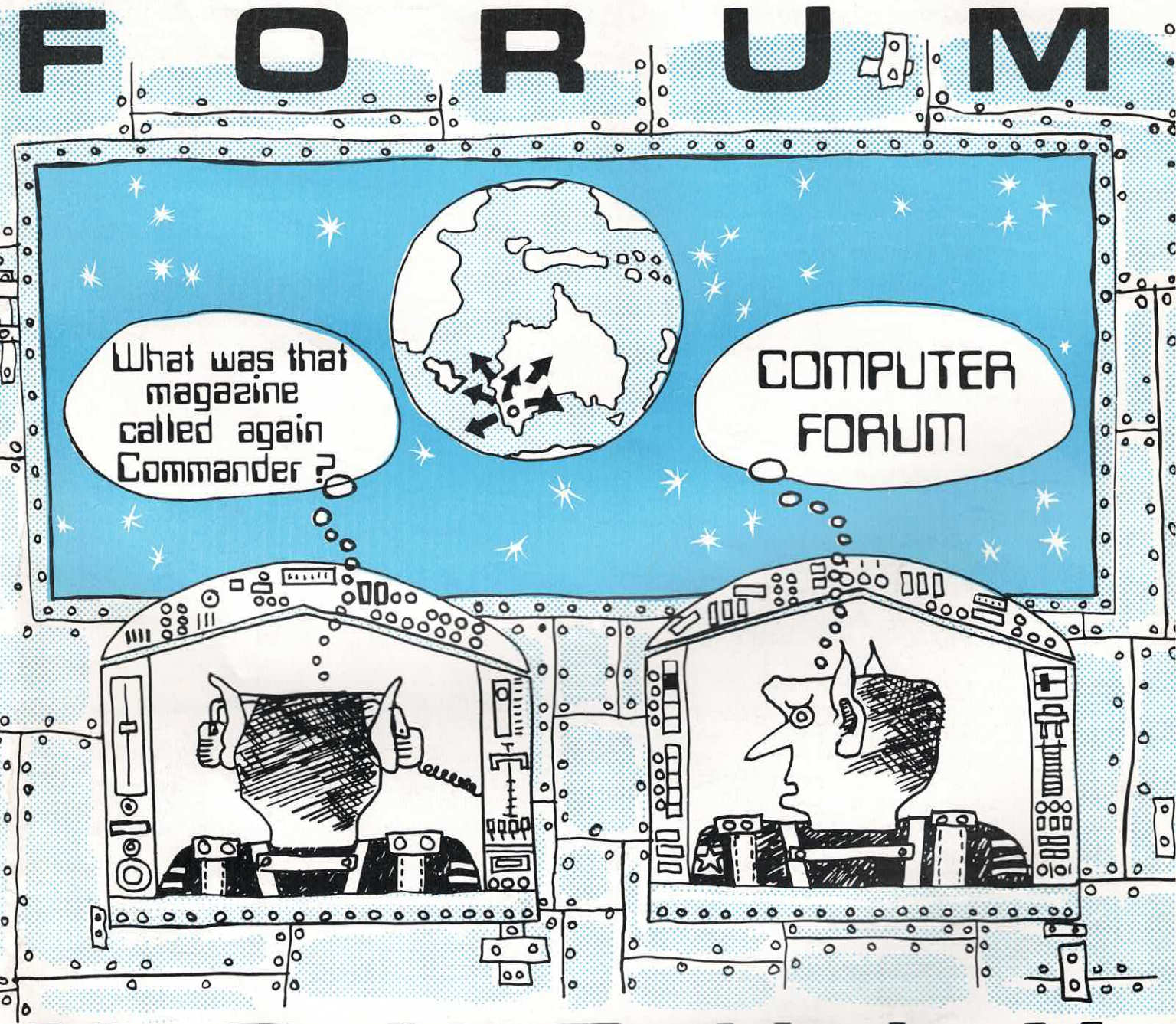
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CONTENTS

ADVENTURES ANONYMOUS.. 2

BEYOND BASIC.....4

PROGRAMMING SPRITES...7

ANOTHER RESET SWICH...9

MSX BASIC TUTORIAL. ..11

MAILBOX.....14

S.W.A.P. 16

HIGH GAME SCORE.....17

LISTINGS.....23

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EDITORIAL

---- MAYBE ----

I think I stirred up quite a bee's nest in my last Editorial by saying that we sold 30 Hit Bit's. I had numerous phone calls from people wanting to by a computer. So for now I will just say we are trying to get hold of some computers but I will not promise anything. I'll keep you informed.

----- SOFTWARE BUYING -----

If you are interested in buying 2 of the most electrifying MSX games that have been produced in a long time let us know. We intend to write to England and get them in they are called **NEMESIS & NEMESIS2** we intend to get in a copy of each to judge how much it will cost and how long it takes to get to us from England. The graphics and music in these programs is absolutely brilliant, a lot of work has gone into making these games. Don't be fooled by the name I would rate these games as being the best I have ever seen. If you are interested in buying one or both of these watch this space because we intend to let you know by the next issue all costs.

------ NEWS ------

Unfortunately we have no news from England or Holland things seem to be very slow at the moment.

------ SOFTWARE SWAPPING ------

All the people who sent in their software lists will receive a list through the mail with all of the available software on it, which will also include the owners Name, Address and Phone No.
All details of swapping will be included on the lists.

C O V E R W I N N E R

This issues cover winner is -

--- Tim A. Jessel of Victoria ---

Very original idea Tim, well done.

ADVENTURES ANONYMOUS



WITH YOUR GUIDE ALAN CRAWFORD

Welcome once again to A.A. The good news this month is that the price of my adventure, THE MYSTERY OF LETHE CASTLE wi emain at $15 and not increase as stated last month. As there is only 999 copies left, then I would suggest you order your copy as soon as possible. As I said last month I am working on my second adventure game, and depending on the interest generated by the first adventure. It too will be available through this magazine, but failing that I may have to look elsewhere, where there is a demand for new and original software.

The popular consensus of opinion is that text adventures appeal to the majority of adventure game enthusiasts as opposed to graphic adventures, which tend to fall down in the respect of vocabulary and overall length. I dont agree fully with this, I think there is a place for both types of adventures, a lot of adventure game players do prefer the shorter more colorful games.

My second adventure will be (or is) a text adventure and also a natural progression towards the more complex adventures that are available from people like LEVEL 9. Writing adventure games in basic, as opposed to machine code , one could not hope to write a game that would stand along side the more complex adventures, but to do what I am doing in my second adventure, which is to write the adventure in two parts, it is possible to come close (a sort of level 7 or 8). Both parts of the game will be on the same tape or disc, which will be like getting two for the price of one. The vocabulary is much larger and also more locations. In part one I have about 50 locations and in part two I expect to have about 65. Making a grand total of about 115 locations, which is a lot in anybodys language, (the average adventure game has between 80 and 100 locations).

Part one is already in the hands of some friends who are puting it through it's paces although I do expect to be making some minor changes. Most of part two is on paper and when that is finished I will be transfering it to the computer, so there is still a lot of work to be done before my second adventure is available.

I obviously cant go into any details about this adventure at this stage except to tell you that I am very pleased with the way it is turning out and I am usually my own worst critic. Apart from the fact that there is over 100 locations another feature is the number of directions that I have put in the game, apart from N, S, E, W, there is also NE, SE, SW, NW, UP and DOWN. All exits from the locations are logical exits eg. If you leave a location by the north exit then you will enter the next location from the south. But more about this game at a later stage.

As I mentioned last month SKULL ISLAND was the last of the DOTSOFT adventures that I had not covered, so now that they have all been done I thought that this month I would do a recap an some of the problems that most people had trouble with and then I will be moving on to LEVEL 9 adventures.

BASTOW MANOR.

THE GATE, climb the tree to the south and break the branch then use this to get the iron key from behind the gate.

TRAPDOOR, is under the case in the shed, also open case.

FRONT DOOR, do nothing here, but leave ladder under window to the east.

DOOR WITH BEAM, use the crowbar from the toolbox, remove it.

SNAKE PIT, the branch will bridge it, then south to toolbox.

LIBRARY, wear the armour before going east from foyer.

ARMOUR, in the wine cellar south as far as you can go.

SILVERKEY, in closet in kitchen, then in the mailbox.

KITCHEN, the meat will rid you of the dogs at the end.

STUDY, the silverkey opens the study door, move the lamp to open up the panel and turn wheel to open the treasure room in the stone cell.

KING SOLOMONS MINES.

CAVE, light web with matches to clear the way south.

SCEPTRE, in the lagoon, dive lagoon to find it then use it to move rock and block the stream.

BRIDGE, east onto bridge then run, climb it at other side.

NATIVE, give him the necklace and the box.

DESERT, to cross go east, south and west,carry the canteen.

LAKE, shake palm tree, drop frond and go frond to cross.

STATUE, insert idol to open hole, theres a crystal here too.

CREVISE, give the old man the gem in return for a lamp.

HUT, cut rope and smash hut to escape.

LEDGE, insert sceptre to open up the hole to a cave.

THE MAD MUMMY.

START, open door and go north.

KITCHEN, use the fork from the dining room to pick lock.

STUDY, the matches are in the desk drawer, turn up lamp.

STAIRS, break crate in the basement for timber to mend them.

SARCOPHAGUS, insert knob into hole to move lid.

CHIMNEY, tie cord to arrow and fire arrow up the chimney, climb the cord to reach the roof.

ALTAR, drop the book, ankh, ring, birdcage and fish to open up an exit from inside the chimney.

SNAKES, pour tin of petrol and light it to remove snakes.

ANKH, insert it into door in room af columns, twist it.

ROWBOAT, take it to get the fish.

ANCIENT DOOR, read the writing and say the magic word.

Well that just about it for this month dont forget next month will be the start of the LEVEL 9 games, I am sure you will be looking forward to that. If you have any problems with an adventure game that you are playing, just drop me a line and I will try to help you out or if you have overcome a sticky situation in an adventure please let your fellow adventurers in on it. Dont forget to order you copy of LETHE CASTLE I am sure you will not be disappointed.

till next time, bye for now ALAN.

BEYOND BASIC

by Tony Cruise

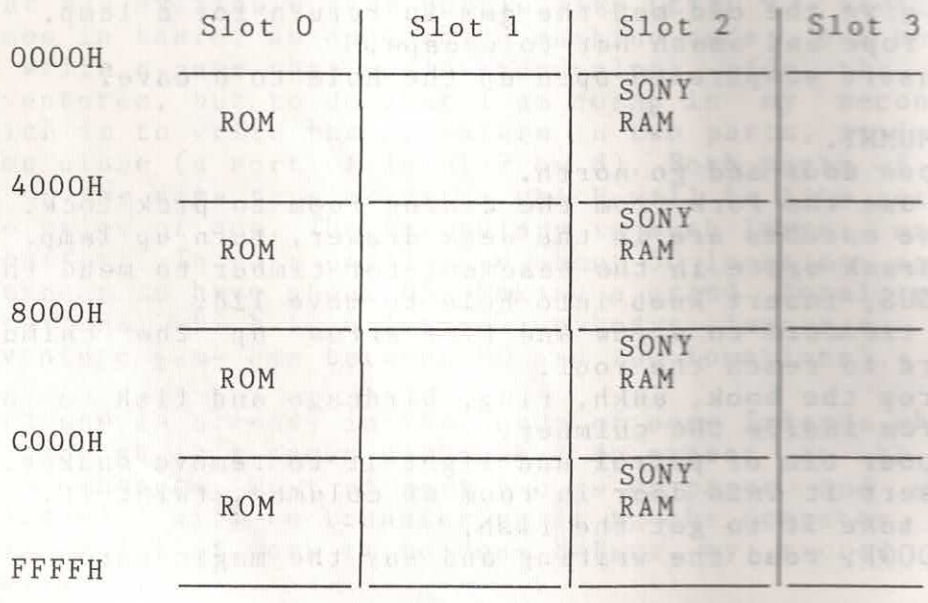
What I aim to do in this series of articles is to provide an insight into machine code programming on your computer. The articles will take for granted that you know how to program in BASIC, but will not require a knowledge of machine code. Full machine code listings will be included for the person who wants to know more, but they will not be necessary for understanding the article.

The series of programs will eventually add extra commands that you can use in your BASIC programs.
The sections will include:-

- Making use of all 64K of your computer's memory
- Printer dump routines
- Screen save and load to memory and/or tape and disk
- Automatic sprite movement with velocities
- Sprite collision detection
- Screen scrolling routines

Part 1 - Using More Memory

A lot of the original MSX game titles that used more than 32K of RAM were very machine dependant. A classic example is "Finders Keepers" which only works on the Sony HIT-BIT. The author took it for granted that all of the MSX machines would have their RAM in the same area. To explain this further look at the diagram below.



Any of the boxes in the picture can hold 16K of RAM or ROM. (Up to 64K can be placed in each box, but that is beyond the scope of this article) Only one of the four slots can be activated for each page of memory. Thus when you turn on the computer the BASIC ROM is activated, so only the top 32K of RAM can be accessed. This is why you only get 28K of RAM free for your BASIC programs. (The other 4K of RAM is used by BASIC as a storage space for system variables eg: cursor position)

MACHINE CODE PROGRAM

```
         1 ORIGIN     ORG 08400H
         2 ;
         3 ; PROGRAM TO FIND EXTRA RAM IN MSX COMPUTERS
         4 ;
F3       5 START      DI                ; Disables all interupts
DBA8     6            IN A,(0A8H)       ; Get page settings
4F       7            LD C,A            ; Store in C
0603     8            LD B,3            ; Loop counter
210000   9            LD HL,0           ; Memory Location to test
C5      10 LP1        PUSH BC           ; Save BC
79      11            LD A,C            ;
B0      12            OR B              ;
CB20    13            SLA B             ;
CB20    14            SLA B             ;
B0      15            OR B              ; Calculate slot
D3AB    16            OUT (0A8H),A      ; Send Bank Information
C1      17            POP BC            ; Restore BC
3E01    18            LD A,1            ;
77      19            LD (HL),A         ;
7E      20            LD A,(HL)         ;
3D      21            DEC A             ;
2802    22            JR Z,EXIT         ; Is it RAM
10EC    23            DJNZ LP1          ; Do loop again
78      24 EXIT       LD A,B            ;
B1      25            OR C              ;
CB20    26            SLA B             ;
CB20    27            SLA B             ;
B0      28            OR B              ; Calculate slot
32E7FF29              LD (0FFE7H),A     ; Store in 0FFE7H
79      30            LD A,C            ; Set pages to normal
D3AB    31            OUT (0A8H),A      ;
FB      32            EI                ; Enable interupts
C9      33            RET               ; Return to Basic
```

BASIC LOADER

```
10 CLS:CLEAR 200,&H000:DEFINTA-Z:A=&HD000
20 READ A$:IF A$ <> "@" THEN POKE A, VAL("&H"+A$):A=A+1:GOTO 20
30 PRINT"  INSERT TAPE/DISK TO SAVE PROGRAM"
40 PRINT"        AND PRESS ANT KEY"
50 A$=INPUT$(1):PRINT:PRINT "  SAVING ....."
60 BSAVE"CAS:RAMSEL",&HD000,A-1:REM Remove CAS: if using disk
100 DATA F3,DB,A8,4F,06,03,21,00,00,C5
110 DATA 79,B0,CB,20,CB,20,B0,D3,A8,C1
120 DATA 3E,01,77,7E,3D,28,02,10,EC,78
130 DATA B1,CB,20,CB,20,B0,32,E7,FF,79
140 DATA D3,A8,FB,C9
150 DATA @
```

Changing which slot allocated to each page is easy. All you have to do is send the correct value to output A8H. The format for the value to send is:-

```
Bit 0 & 1 = Page 0      Bit 2 & 3 = Page 1
Bit 4 & 5 = Page 2      Bit 6 & 7 = Page 3
```

Each two bits can be the following:-

00 - Slot 0, 01 - Slot 1, 10 - Slot 2, 11 - Slot 3

As an example, here is the original setting for the Sony HIT-BIT computer.

```
10 10     00 00 = 160
 RAM       ROM
Slot 2    Slot 0
```

So changing the HIT-BIT to have all 64K of RAM enable the value would be.

```
10101010 = 170          All Pages set to slot 2
```

If you did this in BASIC using OUT(&HA8),170 you would lock up the computer as it suddenly would not be able to find its ROM. Thus it is really only possible to access all 64K of Ram in machine code.
The values I gave in the above examples are for the Sony HIT-BIT and will not work on very many of the other brands. The Sony HIT-BIT has all of its RAM in Slot 2 where as the Spectravideo 728 has their RAM in slot 1. The Yamaha, Sanyo and Toshiba all have their RAM in different memory slots. The following machine code program finds where all the RAM sections are and puts the appropriate value (To enable all 64K at once) in an unused section of memory. The program is completely relocatable ie: it can run in any area of memory. Also you only need to run it once, so it is best left as a separate program.

To use the program to fix a game that does not work on your computer eg: Finders Keepers, set the program up so the slot selection program runs first. Then replace the slot selection routines with a short piece of code like:-

```
F3          1 DI
DBA8        2 IN A, (0A8H)
F5          3 PUSH AF
32E7FF      4 LD A,(0FFE7H)
E63F        5 AND 00111111B  ; Use only the top 16K RAM
D3A8        6 OUT (0A8H),A   ; Send value
:
:   PUT THE CODE TO TRANSFER MEMORY HERE
:
F1          7 POP AF         ; Restore AF
D3A8        8 OUT (0A8H),A   ; Send value
FB          9 EI             ; Enable interupts
C9         10 RET            ; Return to Basic
```

Thats all for this section, next issue I will cover using the extra memory for variable storage..

PROGRAMMING WITH SPRITES - PART 4.

by Lucille Parker..

SPRITE ANIMATION

I think the most interesting and challenging area of programming with sprites is in trying to achieve the effect of animation. This is done by quickly alternating the appearence of two or more sprites on top of each other and at the same time moving them around the screen.

This brings up the problem of removing sprites from the screen. The Reference Manual says to put the Y co-ordinate of your sprite to 209, however any location just above or below the screen will do. If you put the Y co-ordinate of a sprite to the magic number 208, all sprites from that sprite's plane back to plane 31 will disappear. This is an easy way of getting rid of a lot of sprites at once. Another way of hiding sprites without actually removing them is to change their colour to 0 zero (transparent).

Multi-colour sprites can be created by making several small sprites in different colours and placing them on the screen close to each other or even overlapping. They can then be moved around the screen as one object.

My demonstration program this month shows you how to create a little munching "pacman" figure which moves around under joystick control.

```
10 ' PACSPRITES
20 COLOR 2,1,1:SCREEN 1,0:REM SCREEN 2,0 FOR MSX
30 FOR A=1TO8:FOR B=1TO8
40 READ C:D$(A)=D$(A)+CHR$(C):NEXT B
50 NEXT A
60 DATA 52,118,247,215,255,255,126,60
70 DATA 66,227,247,223,255,255,126,60
80 DATA 60,126,247,255,240,255,126,60
90 DATA 62,127,246,252,248,252,127,62
100 DATA 60,126,255,255,235,239,110,44
110 DATA 60,126,255,255,251,239,199,66
120 DATA 60,126,239,255,15,255,126,60
130 DATA 124,254,111,63,31,63,254,124
140 FOR T=1TO7 STEP 2:READ W(T),Z(T):NEXT
150 DATA 0,-2,2,0,0,2,-2,0
160 SPRITE$(1)=D$(3):SPRITE$(2)=D$(4)
170 X=120:Y=90:C1=2:C2=0
180 ST=STICK(0)+STICK(1)+STICK(2)
190 IF ST=0 OR ST=2 OR ST=4 OR ST=6 OR ST=8 THEN 210
200 IF ST THEN X=X+W(ST):Y=Y+Z(ST):SPRITE$(1)=D$(ST):
SPRITE$(2)=D$(ST+1)
210 PUT SPRITE 1,(X,Y),C1,1:PUT SPRITE 2,(X,Y),C2,2
220 FOR T=1TO50:NEXT
230 SWAP C1,C2
240 GOTO 180
```

Lines 30-50 read in the sprite data for 8 sprites, 2 for each direction, and store the information in D$( ).
Lines 140-150 the information being read into W(T) and Z(T) will be used for updating the X and Y co-ordinates of these sprites.
Line 160 initial sprite patterns for sprites 1 and 2.
Line 170 starting position for sprites and colour numbers.
Line 190 prevents mysterious happenings. You must account for all possible states of the joystick.
Line 200 if joystick is pressed, X and Y co-ordinates are updated. Sprite patterns are changed according to the direction of the joystick.
line 210 puts both sprites on the screen on top of each other. One will be green and the other transparent.
Line 220 time delay.
Line 230 swap sprite colours to give the effect of animation.

I hope you have found this series of articles useful, and are now on your way to writting some terrific programs to do justice to the capabilities of our wonderful computers.

- - - - - - - - - - - - - - - -

We would like to thank Lucille for her fine effort in showing us what can be done with sprites and I'm sure we have all gained a little more knowledge from her column and appreciate the work involved in puting it all together.

In our next issue we have another look at sprites as well as various commands we may use in programming, this column comes to us curtesy of KRANK KEAN of New South Wales.

ANOTHER RESET SWITCH FOR THE X'PRESS

A simple mod to the SVI X'press is a reset button. The button can be located under the drive on the right hand side of the case.

Two wires are required - one to the right hand end of the diode D1 under the drive (to the left of IC7 ..ED) and the other wire from the switch has to be connected to ground or earth. This can be connected under the hold down screw for the disk drive BRACKET (extreme right hand corner). These two wires need to be connected to a normally open, push button switch such as a C & K 8121 or similer. Anyone that is reasonably adept with a small soldering iron could do this usefull modification as there is no high voltage in the cabinet.

I'm trying out some other modifications, at the moment, and if they work out successfully, I'll write them up

NB. A caution to the people undertaking the above modification. Be carefull not to bend the multi-conductor ribbon cable connecting the keyboard and the main board to much, as it is not super flexible.

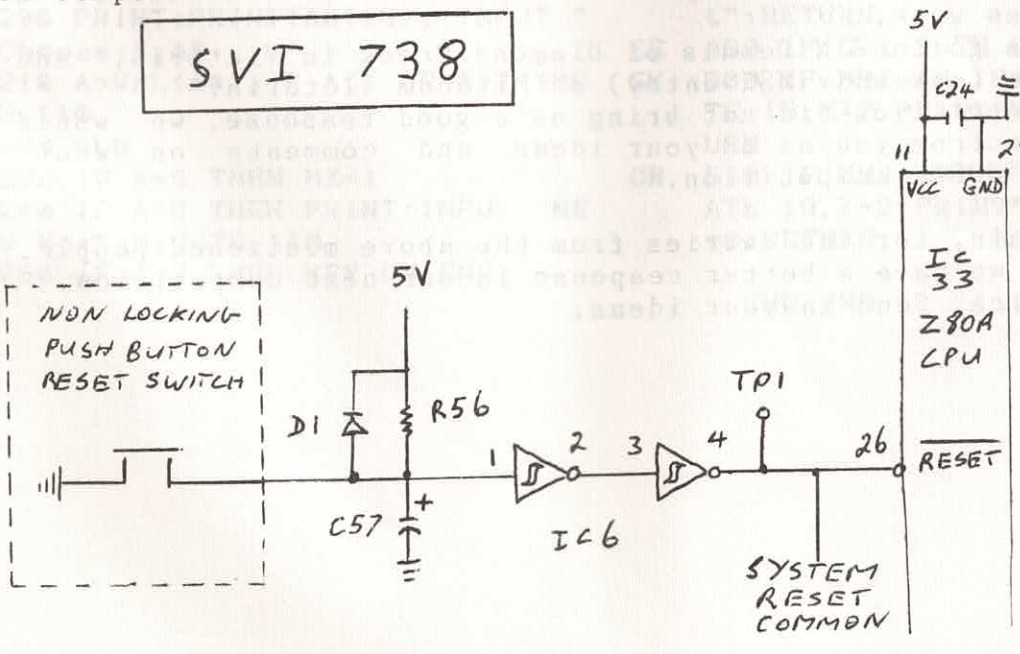
Regards Pete Smith, Bell Post Hill, Nth Geelong

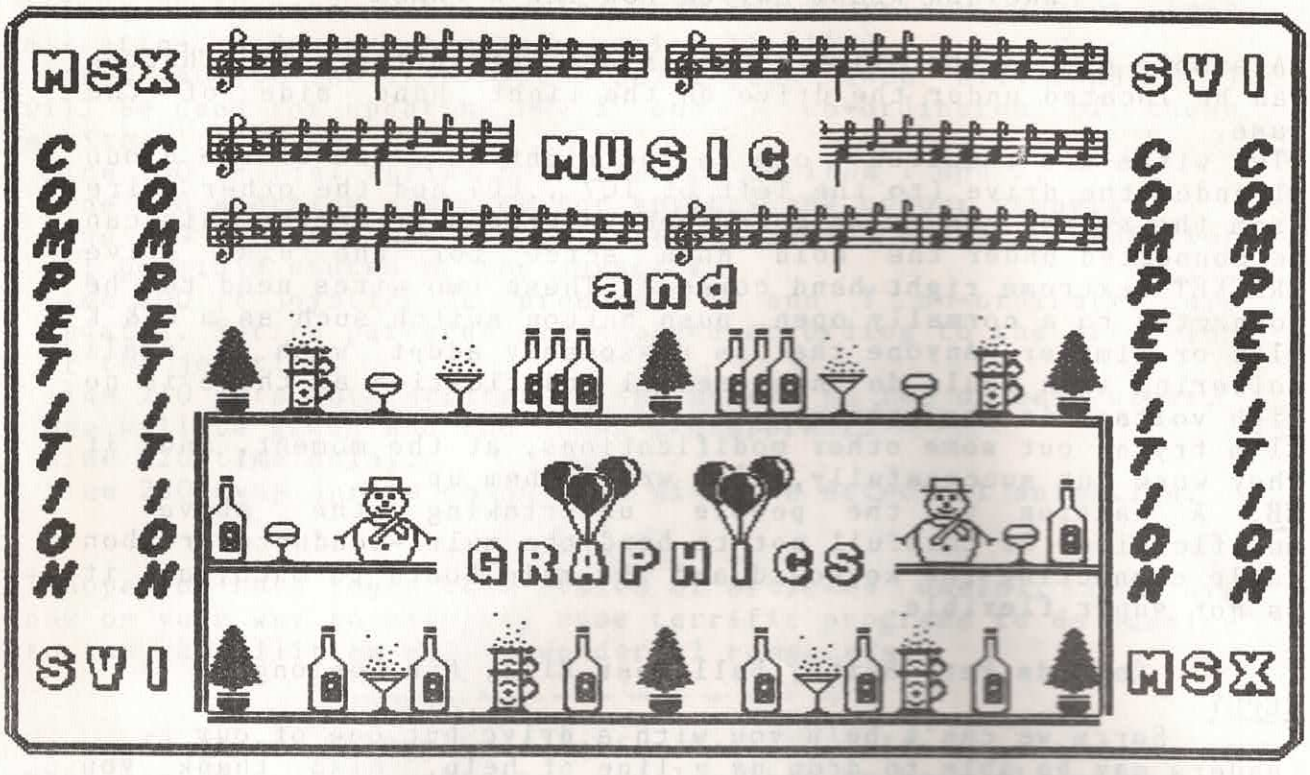
REPLY

Sorry we can't help you with a drive but one of our readers may be able to drop us a line of help, also thank you for the mod, Pete, and we would very much like to hear how your other projects go. Another thing for anyone doing the modification mentioned here, as MSX International Users club say, this will void any and all guaranties.

Also, becarefull of which side of the diode D1 you place your wire. One side goes to the 5 volt rail and the other side links up with the +ve side of capacitor C57, one leg of resistor R56 as well as pin 1 of IC6 (this is the side you want). The output from IC6 is on pin 4 which connects to IC33 pin 26. IC33 is the Z80 chip, and pin 26 is the reset leg of said chip.

For those that are like me, chicken, you can also use the cartridge connection to the main board (away from any chance of damaging any chips with to much heat). Pin 15 is the reset leg and pins 41 + 43 are ground (earth). This is the same for ALL MSX computers.





Well here it is, what you have all been waiting for. The results of the MUSIC and GRAPHICS competition. Due to the **GI-NOR-MOUSE** responce we had initialy, we left it open a bit longer, and sure enough, the number of entries doubled. Thats right, it went from two (2) all the way to four (4) entries. We had three (3) from the SVI owners, and one (1) from the MSX side. (To say that we are disappointed is an understatement.)

Hank Zuydam, the Secretary of SUSPECT (the Sydney SVI users group), is the winner of the **GRAPHICS plus MUSIC** section with his program called CHRISTMAS.

The winner in the **MUSIC** division is PETER WHITE of Birchgrove N.S.W. with his entry of ALLIGATOR. Peter also submitted a couple of other programs in this division.

Your cheques are "IN THE MAIL" and should be with you soon. For their troubles, Hank picks up $40.00 and Peter $25.00. All for a "FEW" minutes work.

Also thanks go to P N Dennis of Diamond Creek in Victoria, and Ian Davidson (our only MSX entry) of Mitcham Victoria.

As this competition did not bring us a good response, we would like to hear from you as to your ideas and comments on what would be a "GOOD" competition.

Thanks, again, for the entries from the above mentioned people. And we hope we have a better response in our next competition, of your choice. Send in your ideas.

SUMS

```
CB  10 CLS:KEY OFF
DH  20 RDRST=RND(-TIME)
JG  30 PRINT:PRINT"You must enter a
    figure from 1 to 99"
PE  40 PRINT:PRINT"For example - If
    your child was 6 you
DH  50 PRINT"would enter a number u
    p to about 9"
HL  60 PRINT"What this means is no
    matter which typeof sum is chos
    en, EG- (ADDITION) then  the hi
    ghest that sum could be is 9+9"

KH  70 PRINT:PRINT"It is up to you
    how difficult you make the sums
    for your child."
FF  80 PRINT
IL  90 PRINT:INPUT "Enter a number
    between 1 and 99";N
DJ  100 IF N<1 THEN RUN
CD  110 CLS:LOCATE 9,3:PRINT"ARITHM
    ETIC PROBLEMS"
CA  120 PRINT"
              No:";N
BP  130 PRINT:PRINT TAB(12);"1. ADD
    ITION"
AN  140 PRINT:PRINT TAB(12);"2. SUB
    TRACTION"
BJ  150 PRINT:PRINT TAB(12);"3. MUL
    TIPLICATION"
DE  160 PRINT:PRINT TAB(12);"4. DIV
    ISION"
CP  170 PRINT:PRINT TAB(12);"5. MIX
    TURE "
AM  180 PRINT:PRINT TAB(12);"6. CHA
    NGE NUMBER"
BO  190 PRINT:PRINT TAB(12);"7. END
    "
CE  200 PRINT:PRINTTAB(15);:INPUT "
    Choose:";A$
EN  210 A=VAL(A$):IF A<1 OR A>7 THE
    N 110
BK  220 CLS
AJ  230 IF A=5 THEN MX=1
JO  240 IF A=6 THEN PRINT:INPUT "NE
    W No:";N:GOTO 110
FB  250 IF A=7 THEN KEY ON:END
CJ  260 FOR X=1 TO 10
    :LOCATE 20,1:PRINT"Wrong:";WR
FF  280 IF MX=1 THEN A=INT(RND(1)*4
    )+1
CJ  290 N1=INT(RND(1)*N)+1
DE  300 N2=INT(RND(1)*N)+1
BF  310 IF A=1 THEN AN=N1+N2:LOCATE
     4,X+2:PRINT USING "##";N1;:PRI

    NT" + ";:PRINT USING "##";N2;:P
    RINT" =";:INPUTYA$:IFYA$=""THEN
     310 ELSE MM=VAL(YA$):GOSUB 470

AK  320 '
BD  330 IF A=2 THEN AN=N1-N2:LOCATE
     4,X+2:PRINT N1"-";N2;"=";:INPU
    TMM:GOSUB 470
AI  340 '
AJ  350 IF A=3 THEN AN=N1*N2:LOCATE
     4,X+2:PRINTN1;"x";N2;"=";:INPU
    TMM:GOSUB 470
AG  360 '
AG  370 IF A=4 THEN GOSUB 420
FO  380 IF A=4 THEN LOCATE 4,X+2:PR
    INTN1;"÷";N2;"=";:INPUTMM:GOSUB
     470
DL  390 NEXT X
CJ  400 PRINT:PRINT:INPUT "Press en
    ter to continue";ZZ:GOTO 110
AI  410 END
CN  420 IF N2>N1 THEN N1=N1+1:GOTO
    420
AC  430 N1=N1+N1
CE  440 AN%=N1/N2:G=1
AH  450 IF AN%*N2=N1 THEN 460 ELSE
    N1=N1-1:GOTO 450
BO  460 RETURN
AG  470 '
GD  480 IF MM=AN OR MM=AN% THEN CO=
    CO+1:LOCATE 19,X+2:PRINT"Correc
    t":RETURN
CE  490 IF G=1 THEN 510
GK  500 IF MM<>AN THEN WR=WR+1:LOCA
    TE 19,X+2:PRINT"Wrong :";AN:RET
    URN
CB  510 IF MM<>AN% THEN WR=WR+1:LOC
    ATE 19,X+2:PRINT"Wrong :";AN%:G
    =0:RETURN
```

THE END

MSX BASIC TUTORIAL

PART 6

I was asked some months ago to write a small educational program that deals with mathematics, well here it is.

I also thought why should I do all of the programming. So now it is your turn to figure a few things out.
There is a BUG in the program, when you pick a particular option and then return to the main menu to choose another option, it doesn't let you, WHY???. Just a little hint MX=1

There are a few other little things but as we break up the program I will mention them.

Once again I will only explain the things I think you need to know or expand on things you already know.

```
 20 RDRST=RND(-TIME)
```
You may recall this is used to get a RaNDom number by using the internal clock.

```
 90 INPUT--- ;N
```
This line is what stores your number under N.

```
120 PRINT " SPACE SPACE SPACE ETC No:";N
```
There must be a faster way of doing this line rather than using a pile of spaces.

```
130 - 190 MAIN MENU
```

```
200 INPUT"Choose:";A$
```
How could we make this line work as soon as you push any number between 1 & 7, there are 2 ways of doing it

```
210 A=VAL(A$) etc
```
This is where we convert a string ($) into a numeric value and store under a variable called A, then we check it to make sure it is not less than(<)1 or if it is greater than(>)7.

```
230 IF A=5 THEN MX=1 (the bug starter)
```
This line sets MX to equal 1 so it can do line 280.

```
260 FOR X=1 TO 10
```
Starts up a loop for 10 sums

```
280 IF MX=1 THEN A=INT(RND(1)*4)+1
```
If MX=1 this line will always come into use.
It sets up a RND number between 1 & 4 and stores it under A.

```
290 N1=INT(RND(1)*N)+1
```
Stores a number under N1, it gets it's number from what ever number you choose in the beginning of the program and picks a number between 1 & (N) your number

```
300 N2=INT(RND(1)*N)+1
```
Same as 290 but stores it under a different name.

310 IF A=1 THEN etc...
In other words, if your option equals No 1 (addition) then the (AN)Answer = No stored in (N1) + the No stored in (N2):Locate 4 (across),X+2(down):Print USING "##";N1 (The ## (HASH)symbols are use to print blank spaces, so if the No is 1 or the number is 100 they will keep them in a straight column format).

330 IF A=2 THEN etc...
This is where you get to do a bit of programmin, all you have to do is put the print out on the screen in column format as I did in the previous line.

380 IF A=4 THEN etc...
You will notice on this line there is a division sign (÷) that can be printed on the screen by pressing 3 keys, hold down [SHIFT],[GRAPH] and press [?].

420 IF N2>N1 THEN etc...
If the RND No chosen by N2 is greater(>) than the No chosen by (N1) then keep adding 1 to (N1) until it is not.

440 AN%=N1/N2:G=1
The (%) figure after the (AN) rounds off the number-
Try this next time your computer is in basic-

A%=6.99	A=6.99
B%=3.22	B=3.22
PRINT A%/B%	PRINT A/B
ANSWER 2	ANSWER 2.1708074534161

As you can see the % figure rounds off the number.
The rest of the line divides the first No(N1) by the second No(N2) and then sets G to 1.

450 IF AN%*N2=N1 THEN etc...
This line checks to see if the answer (AN%) times (N2) = (N1), this means it can be divided into.

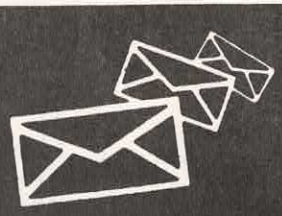
480 - 510
These lines are used to check if you got your answer correct or incorrect, and prints up the appropriate message.

I have been asked to write a small Word Processing program for my next column, I will try to keep it as small as possible with as many features as you need.

If you have an idea for a program you would like to see in the magazine send it in and I will see what I can do.
Also if you have written a program or you are writting one and you may be having some problems, send it to me and include enough postage for return mail and I will try to fix it for you or at least set you in the right direction.

Until next time HAPPY PROGRAMMING.....

MAILBOX



Here is a letter from Bernard Waslin of Victoria.

If you remember when I got the Disk Drive, Controller & Expander from you, I was having some difficulty joining the SV Keyboard and Expander, and you were kind enough to send me a small 'WIRE-TAPE' to join the two, (with sockets on both ends).

Well I have noticed that the same tape will go into the cartridge slot 'B' at the rear of the Sony Hit-Bit, although it is for cartridges. I was wondering if (with the wire tape) I could join the Hit-Bit with the SV Expander to use the disk drive..... the reason I ask, is that the slot 'B' on the Hit-Bit is 'INPUT' (cartridge) and not 'OUTPUT' and I will not try to connect them together in case I mess things up.

REPLY

Bernard did send in a S.A.E. for a quick reply, but unfortunately we could not help him. I do not know anyone that has tried this, but, if any of our readers can help please write in and we can all benefit.

We also have a question from Peter Smith of Victoria.

I have a question for the MAG.
I have a SVI 757 MSX232 cartridge supplied with software on a 5 1/4in disc. Does anyone know what drive format is used for the disk, as when I try to run it on my second drive also 5 1/4, the logo comes up but wont run.

Through a special CPM program, I have found out it should run on a SVI 728 MSX computer, which I think uses a SVI 707 5 1/4in drive.
Can anyone enlighten me. How many tracks per side & is it single or double sided.

REPLY

You are correct in the fact that the SVI 728 with a 707 drive will run the software it was a one of a kind type of software.
It is also set at Double Sided 40 track.
If anyone else can help please drop us a line.

WANTED WANTED WANTED WANTED WANTED

Can someone send the instructions for Knightshade & Knightlore to me. I have had the programs for about 2 weeks and I have been trying to figure them out, I don't know who sent the programs to me, but I have no instructions.
They look like very good games, I only wish I new what I was doing.
Somebody pleeeeeeeeeease help.....

Ross the frustrated editor and partime game player.

Dear Sir,

I have a little information to pass on - so I thought I'd make use of the opportunity to ask another couple of questions, while I'm at it.

Greensborough Computers does indeed have (at the time of my enquiry to them), the Yamaha FD-03 Single Sided Disk Drive (although I know not how many) at $599. Also Yamaha Music Centre in Sydney has one only of the double sided variety remaining, at $999.

It does seem, to the technically uninitiated such as my self, that the more enquires one makes, the more bewilderingly complicated the whole affair becomes!

Due to the fact that I want to use my Bank Street Writer word processor cartridge in conjunction with the disc drive - and that my computer has only one cartridge slot - I am trying to find out about alternative methods of connecting the disc drive.

If one were to wire the disk drive cartridge in parallel with the computer's cartridge slot - thus leaving it physically free to accept a programme cartridge - would such a system work?

Can one get some sort of adaptor plug to go between the expansion bus and the disc drive's cartridge? - or is an interface unit with other componentry necessary? If I knew the pin-outs of my computer's expantion bus, and the corresponding connections on the disc drive's cartridge, could I simply connect them directly? If so, would this set up allow the drive to properly interact with the programme cartridge?

I would have thought it commonsense to connect a disc drive to the expantion bus, leaving the cartridge port free. Why do the manufacturers make them to plug into the cartridge port?

Thank you for your time, yours faithfully,
Allen Bird. Bomaderry, N.S.W.

REPLY

Thank you for the information on where to get disk drives and how much they cost.

As to your questions on the different ways in which to connect your disk drive cartridge, they all "MAY" work. The only one that may give some trouble is connecting it to the cartridge slot. Most of the MSX range have two cartrige slots, one to act as a cartridge slot and the other to act as an expansion port. These two "Cartridge" ports do not usually share the same SLOT.

EG. the Sanyo PHC-33 has two cartridge ports situated on the top. The front port is set up to be in SLOT 1 and the back port is in SLOT 2. In the case of the SVI 738, which has a built-in disk drive and only one cartridge port, it is set up to appear in SLOT 2.

So wiring the disk drive cartridge to the cartridge port may cause the computer some confusion when trying to run a cartridge and the disk drive.

I have heard of a cartridge adaptor that plugs into the rear of the Toshiba, but I have not seen it. It was advertised in the old MSX Computing mag. (WV Supplies, Bank Buildings, Bank St, Newton Abbot, Devon, TQ122JL, England. Price was #14.95)

The pin-outs for the expansion bus are the same as the cartridge port.

The fact that 'most' of the MSX computers have two cartridge ports is the most probable reason for the manufacturers making the disk drive interface in the form of a cartridge.

I hope this has been of some help to anyone with the same questions. If anyone knows were an adaptor is available in Australia, then please drop us a line.

Swapping Words And People

HELLO, My name is Noel Bergman and as Sectretary/Librarian for the Brisbane Spectravideo and MSX Users Group, I thought it was high time the hard working staff members who put COMPUTER FORUM together got some feed back.

As secretary for our club, I have made contact with several other SVI/MSX User groups around the world and it is from these sources this column will gather it's items of (I hope) interest and news.

As you can see from the title of this column, we will be swapping news and comments from the different User Groups we presently correspond with, and others as contact is made. Hopefully this will create more interest in our excellent machines, different User Groups and Australia's only SVI & MSX magazine.

CANADA
The Canadian group is named SMUG of MONTREAL, with over 60 members and knowing how great our computers are, they certainly have something to be smug about. They produce an excellent monthly news letter which is Bi-lingual, French and English. They have adopted an interesting approach to the falling membership syndrome by opening up their membership to include IBM, TEXAS INSTRUMENT and BONDWELL computer owners. The editor is in the process of compiling a dictionary of computer terms, as a result of their member requests, which points to one aspect of computing books which is lacking.

SOUTH AFRICA
This group is called GREENSTAR NATIONAL COMPUTER CLUB and their Bi-monthly magazine is titled COMPUMAG. It also is of excellent quality and extends up to 50 pages, packed with everything from program listings to abstract maths problems and answers. To give you some idea how big SVI/MSX is in South Africa, their membership totals 8300 of which 60% is SVI, 35% is MSX and 5% is IBM. They write, produce and sell some of their members programs to supplement imports of commercially produced programs, we should get some good ideas from this group.

NEW ZEALAND
The SPECTRAVIDEO USERS GROUP of WELLINGTON also produce a Bi-montly magazine, and one aspect which impressed me about this active group is that up to recently they took part in a country wide computer broadcast service. This only stopped when other brand computer groups pulled out and made it uneconomical to continue. Also they devoted a considerable portion of their news letter to information/listings of other computer languages which can be run on our computers, such as Forth & Pascal.

BRISBANE
Our club boasts membership from just about every state in Australia, and it is very interesting to note that some of our members like computing so much that they own several computers. It's lucky they have ten fingers to use all those computers. Our membership totals 38, But, despite the small number, the club benifits from the great enthusiasm of it's members.

In future I will include some interesting snippets from the various newsletters we receive. Bye for now NOEL...



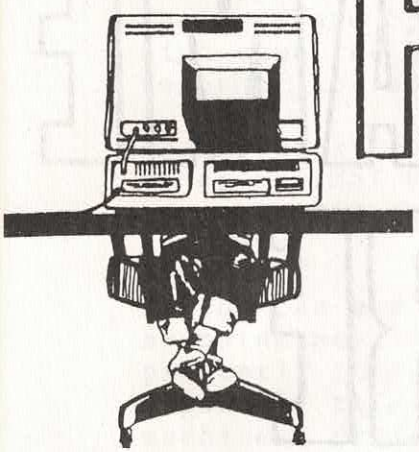
HIGH GAME SCORE

ALL GAMES ARE SVI MODE.

GAME	NAME	SCORE	LEVEL
BOA	MIKE ZUYDAM	1,980	
CITY BOMBER	MIKE ZUYDAM	1,250	
CROSSFORCE	COLIN KERIM	426,000	
FRANTIC FREDDY	MIKE ZUYDAM	31,000	
GHOSTTRAP	HANK ZUYDAM	19,000	
JUMPING JACK	HANK ZUYDAM	2,960	
KUNG FU	MIKE ZUYDAM	31,010	
OLD MAC FARMER	MIKE ZUYDAM	24,000	
PACMAN	JOHN ZUYDAM	18,500	
ROADFROG	MIKE ZUYDAM	130	
SASA	JOHN ZUYDAM	22,650	
SPECTRON	JOHN ZUYDAM	6,650	
SWINGMAN	HANK ZUYDAM	149	
TURBOAT	MIKE ZUYDAM	23,800	
LEUCOCITE	MIKE ZUYDAM	8,780	
TILER	JOHN ZUYDAM	1,636	
CASPER	MIKE ZUYDAM	2,850	
CARSLICK	ZAC ZUYDAM	636	
JOUST	JOHN ZUYDAM	1,050	

ALL GAMES USE THE COLICO ADAPTER FOR THE SVI

GAME	NAME	SCORE	LEVEL
DONKEY KONG JNR	HANK ZUYDAM	15,700	
LOOPING	MIKE ZUYDAM	11,140	
VENTURE	HANK ZUYDAM	633,000	
LADY BUG	MIKE ZUYDAM	60,000	
MR DO	HANK ZUYDAM	104,300	
PEPPER II	MIKE ZUYDAM	215,540	
CABBAGE PATCH KID	ZAC ZUYDAM	42,600	
FRENZY	MIKE ZUYDAM	9,386	
Q-BERT	JOHN ZUYDAM	104,190	
MINER 2049ER	JOHN ZUYDAM	12,500	
TANK WARS	HANK ZUYDAM	38,500	
BURGER TIME	MIKE ZUYDAM	65,500	
MOUSE TRAP	JOHN ZUYDAM	100,980	
ZAXXON	MIKE ZUYDAM	32,600	
TIME PILOT	HANK ZUYDAM	17,100	
DONKEY KONG	MIKE ZUYDAM	54,900	
SMURF	MIKE ZUYDAM	5,800	
SQUISH-EM-SAM	HANK ZUYDAM	29,420	
MONTEZUMA	JOHN ZUYDAM	378,650	
SAMMY LIGHTFOOT	HANK ZUYDAM	46,000	



HIGH GAME SCORE

ALL GAMES ARE MSX MODE

GAME	NAME	SCORE	LEVEL
ALI / FORTY TH	PETER COTTER	80,100	
ALPHAROID	ALAN CRAWFORD	379,800	LEVEL 8
ANTARCTIC ADV	DAMON BURN	67,660	STAGE 11
ANTY	JARED HYLAND	122,600	
ATHLETIC LAND	PETER COTTER	324,500	STAGE 25
BATTLESHIP CLAP	JOHN KNIGHT	60,100	
BEAM RIDER	RICHARD JOHNSON	81,112	
BUCK ROGERS	JASON VESPA	149,000	
CHIMACHIMA	ALAN CRAWFORD	29,560	LEVEL 18
CLAPTON 2	JASON VESPA	70,400	
COMPUT BILLIARD	PETER COTTER	114,300	
DUNGEON ADV-	PHILIP BROUWER	LEVEL 3	
DORODON	JOHN KNIGHT	45,540	
EXA INNOVA I.E	SEAN DAVIDSON	75,160	STAGE 7
GOLF	DAMON BURN	5 UNDER	
GREEN BERET	PETER COTTER	116,300	
H.E.R.O.	JOHN KNIGHT	51,235	
HUMPHREY	KAYE BERGMAN	4,020	
HYPER SPORTS	P. QUINN	475,000	
KING VALLEY	PETER COTTER	49,300	
KNIGHT TYME	CRAIG HURSEY	100%	
LE-MANS	MATHEW HYLAND	35,250	
LODE RUNNER	ALAN CRAWFORD	1,122,475	LEVEL 111
MANIC MINER	KEVIN TATE	66,663	
MONKEY ACADEMY	SEAN DAVIDSON	136,550	STAGE 13
MOUSER	JOHN KNIGHT	35,900	
MR WONGS L/LAUN	JARED HYLAND	1,430	
PIN BALL	B. RANKIN	870,450	
PITFALL 2	P.J. CHRISTIE	199,000	
POLAR STAR	R. DUNCAN	210,000	
RAID BUNGELING	P.J. CHRISTIE	112,030	42 MINUTES
ROAD FIGHTER	CRAIG HURSEY	190,235	STAGE 14
RIVER RAID	D. OLSON	56,650	
SCION	JOHN KNIGHT	131,400	
SENJYO	IAN DAVIDSON	122,550	Level 6
SKRAMBLE	DEL BERGMAN	24,370	
SPECTRON	PHILIP BROUWER	9,640	
SUPER COBRA	JARED HYLAND	113,440	
SUPER CROSS FC.	JARED HYLAND	259,000	
SUPER GOLF	PETER COTTER	-1	UNDER PAR
SUPER TENNIS	DAMON BURN	6 - 0	
SWEET ACORN	JASON VESPA	174,380	
TENPIN	LEE BERGMAN	124	
TRACK & FIELD 1	PETER COTTER	704,630	
ZAXXON	JOHN KNIGHT	52,200	

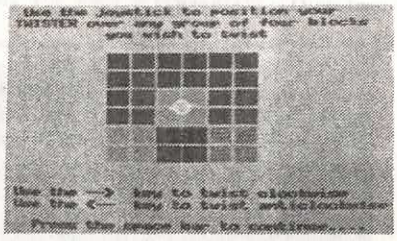
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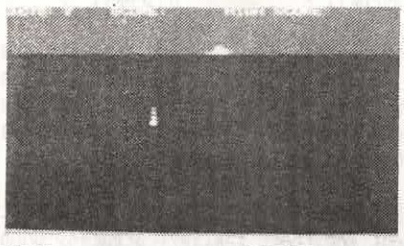
Make cheques payable to COMPUTER FORUM



SVPAK23 - MSXPAK23 -

TWISTER - Like a two dimensional Rubiks cube, this game displays nine different coloured squares each containing four smaller squares. These 36 squares are then jumbled around on the screen and you must rebuild the original display. Its very simple - VERY addictive!

REVERSE - Your task is to arrange the ten jumbled numbers into ascending order by reversing some or all of them. Check it out - its not easy!



TS852 - MSX852 -

GRAND PRIX - You are in the drivers seat in a race to the chequered flag while avoiding slower cars. Hitting another car or two wheels off the track will slow you down and lose points.

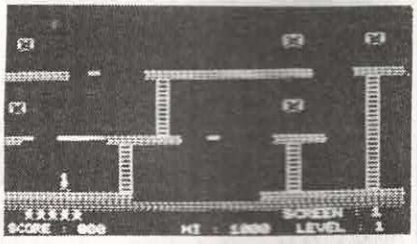
SUBSTRIKE - Drop depth charges on the enemy below. The deeper he is or the faster he moves, the more you will score.



SVPAK24 - MSXPAK24 -

FOX & GEESE - Can you as the Fox penetrate the line of advancing geese and get to the other side of the board before they corner you.

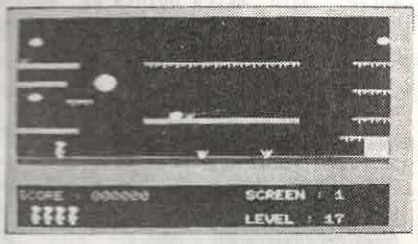
THINK - This is a form of word mastermind. You must guess the word that the computer has selected from the vast number stored in its data banks. This game will certainly make you do as the name suggests.



SVPAK7 - MSXPAK7 -

BOMB SCARE - Very addictive game in which you must climb to different platforms and defuse 5 bombs before your bonus runs out. Has 6 separate screens with the hazards of moving platforms and conveyor belts with holes in them.

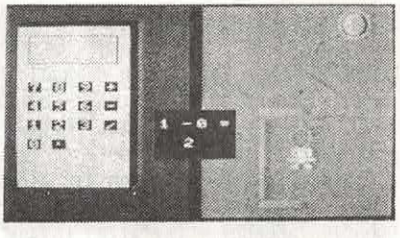
DUNGEON ADVENTURE - A text adventure where you explore the dungeon fighting monsters and gain treasures to discover the location of the great Orb.



SVPAK21 - MSXPAK21 -

GOLDRUSH - Move the miner around the different levels of the mine collecting gold and dodging high explosive charges and falling rocks. The green ledges crumble if you don't get off quickly. When you have collected all of the gold, get to the lift cage for a new screen.

PYRAMID - This is another fascinating text adventure built around the vaults of a pyramid.



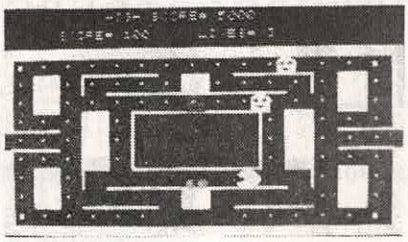
SVPAK25 - MSXPAK25 -

OZMAP - You must drive the car around Australia finding the places required as quickly as you can.

OZQUIZ - Answer general knowledge questions about Australia and her people.

MATH DUEL - Makes practicing maths fun! Has 4 levels of addition and subtraction maths.

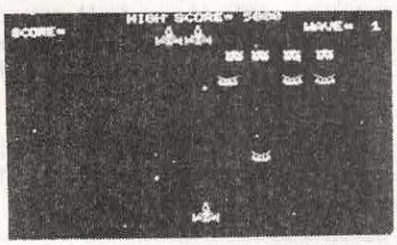
Prices include postage and packing within Australia



SVPAK12 - MSXPAK12 -

MUNCH MAN - Guide your Munchman around the maze eating the dots while avoiding the deadly ghosts. If you eat one of the larger dots you will be able to eat the ghosts for a short while. Gain extra points by eating the fruit that occasionally appears on the screen.

ESCAPE FROM PLANET BETA - This is a text adventure where you are stranded on Planet Beta and must find all the necessary equipment to get off before the planet explodes. Only requires one or two-word commands - eg GET LASER.



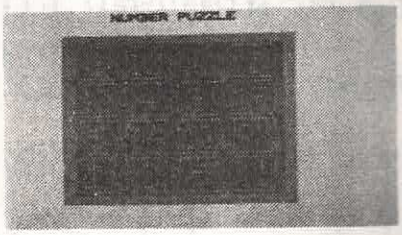
SVPAK5 - MSXPAK5 -

LUNAR LANDER - You are in command of a Lunar Excursion Module with limited fuel but full rotational ability and must land on the lunar surface. Has 8 different landscapes which get harder

GALACTIC ASSAULT - The alien invaders hover above you and one at a time dive on you. Destroy each wave before they get you. For 1 or 2 players.

ROAD RACER - Try and pass as many cars just as fast as you can without crashing.

SPRITE DEFINER - Design sprite shapes on the screen and rotate them when you have finished. Has grids for both sprite sizes.



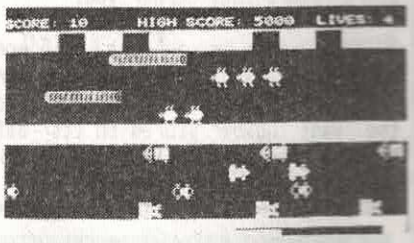
SVPAK20 - MSXPAK20 -

ARTILLERY FIRE - A two player game where each can select a firing angle and velocity to lob shells on his opponent.

NUMBER PUZZLE - Computerised version of the popular hand-held game in which you must slide numbers around into ascending order.

TOWERS OF HANOI - Move all of the discs from peg 1 to peg 3 without placing a large disc on a smaller one.

TENNIS - This is a two player game of Table Tennis.



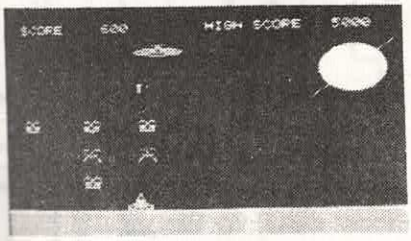
SVPAK6 - MSXPAK6 -

HOPPER - Help Hoppy get across the road (SPLAT!) and then to cross the river (SPLOSH!) and if you are lucky he won't get eaten by the crocs (GULP!!).

SPACE WAR - Challenging game for two players in this duel between your spaceship and that of your opponent. First one to ten points wins.

OTHELLO CHALLENGER - The classic mind game where you must outflank your opponent to capture his pieces. For 1 or 2 players.

PAINTER - Explore the flexibility of your computers colours with this on-screen paint brush.

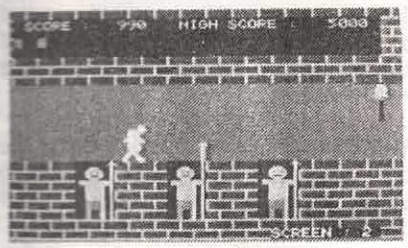


SVPAK9 - MSXPAK9 -

SPACE INVADERS - Shoot the aliens before they invade your planet. Move your ship left and right and shoot missiles at the aliens. Once they have landed the game will end. Score extra points by shooting the UFO which flies overhead.

SUB HUNT - Drop depth charges on the attacking subs and dodge the missiles they fire. You can fire two depth charges at once by firing while moving left or right. Each of the subs is marked with its value.

CHICKEN - Who cares why the chicken crossed the road, we have to find out how! Each time you make it across, the cars speed up.



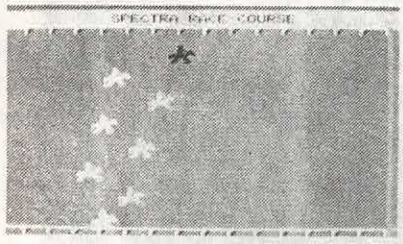
SVPAK11 - MSXPAK11 -

LUNAR RESCUE - Rescue the stranded astronauts from the moons surface one by one. Dodge the asteroids on the way down and destroy them on the way back up for more points.

SKY BASE - Blast the aliens but watch out for their kamikazi attacks. Shoot two sets and then dock with your mother ship. With this added firepower go on to destroy the alien base.

QUASIMODO - Can you as the hunchback Quasimodo rescue the beautiful maiden from the French soldiers. Cross the screen avoiding fireballs and spears, then ring the bell to get the next screen.

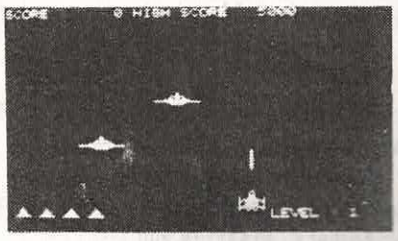
Prices include postage and packing within Australia



SVPAK1 - MSXPAK1 -

SPECTRA DERBY - Place your bets and have a day at the races. Even if you lose its painless.

HANGMAN - The classic word guessing game in which you will be gradually hung for each wrong letter.

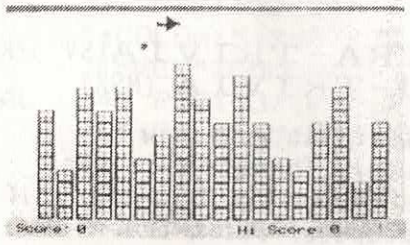


SVPAK10 - MSXPAK10 -

DEMON ATTACK - Shoot the attacking demons as they swoop overhead. You can only destroy a demon when it is fully formed. Killing ten demons gets you on to a harder set.

SEA DIVER - Dodge sea creatures as you make your way to the sea bed. Grab a gold bar and get back to the surface.

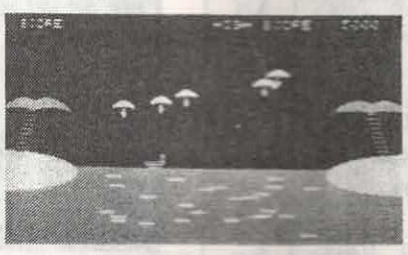
DEATH CITY - Race around the maze trying to shoot your computer opponent but be careful, he is also shooting at you.



SVPAK2 - MSXPAK2 -

SNAKE - Move the snake around to eat the diamonds, but mind you do not cross your own tail.

CITY BOMBER - You must destroy the towering buildings of the city before you can land safely.

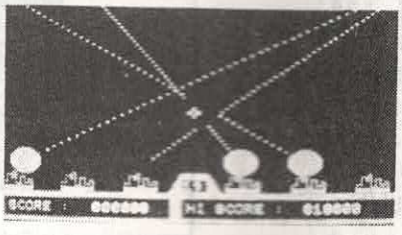


SVPAK8 - MSXPAK8 -

EXPLORER - Watch out for the Pitfalls in this game which will test your control of the joystick. Crocs and other obstacles bar your way as you wander through the jungle in search of treasures.

PARACHUTE - Catch the paratroopers in your boat as they jump from a helicopter into a shark-infested lagoon. The more you catch, the faster they jump.

ANDROID CAVERNS - Move around the caverns of this game and collect the treasure guarded by the androids. But watch out, the androids learn by their mistakes.

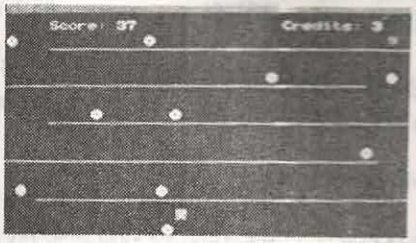


SVPAK22 - MSXPAK22 -

MISSILE COMMAND - Shoot down the incoming warheads before they destroy your cities. You get bonus cities awarded for high scores.

MASTERMIND - Your task is to match the same hidden four-colour combination that the computer has chosen.

LEAKY ROOF - Catch the raindrops leaking through the roof and empty the bucket out of the window before you drown.

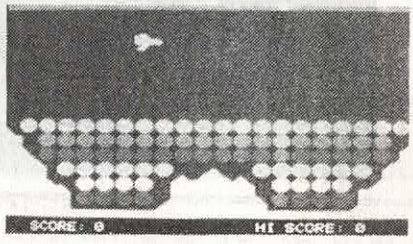


SVPAK4 - MSXPAK4 -

SPECTRA KONG - Win the lady's heart by getting to the top of the screen - but watch out for falling barrels.

TENNIS - A colour version of the original black and white TV game.

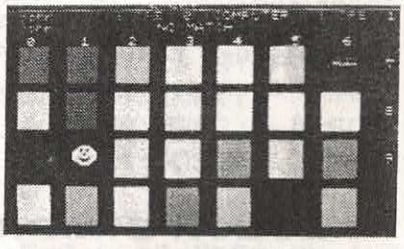
MUSICAL KEYBOARD - This program turns your computer keyboard into a musical keyboard and allows you to select note length and octave.



SVPAK3 - MSXPAK3 -

BOMBS AWAY - Destroy the targets in the ravine below. Be warned - it gets harder the more targets you hit.

BUG CATCHER - Two players move around the garden and the one with the most bugs caught - wins. Great for younger children.



SVEP1 - MSXEP1 -

COPY CAT - The computer will play a set of notes and you must play them back in sequence. There are two levels of play - one note added to the same sequence or a new sequence each time.

MEMORY - A 1 or 2 player game of memory exercise. Uncover two squares on the screen and if the objects revealed match you get another chance

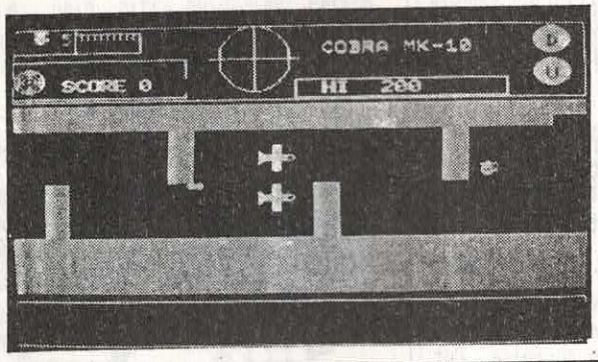
Prices include postage and packing within Australia

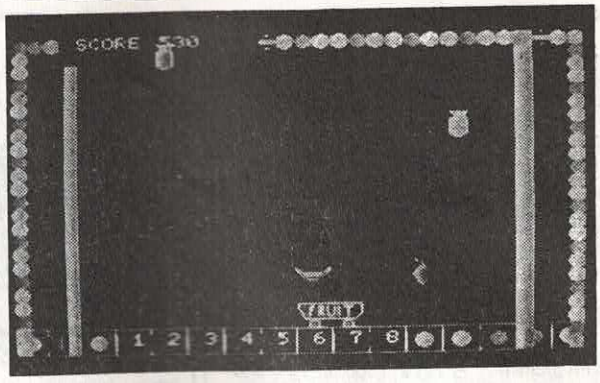
SVPAK26 - MSXPAK26 -

JUMPING COWBOY - A 'platform' game of nine screens where you must collect all the opals to get the next screen.

SUPER SMASHER - Move the bat to smash the ball through the two brick walls. Has three levels of difficulty.

FRUIT CATCHER - Catch falling fruit in your cart. As you catch more, the types of fruit will change and fall at a faster rate.





SVPAK27 - MSXPAK27 -

MIRKWOOD FOREST - Graphics adventure where you must escape from the forest by finding four keys, eating food to maintain energy, while avoiding the spiders. This one will really test your adventure gaming ability.

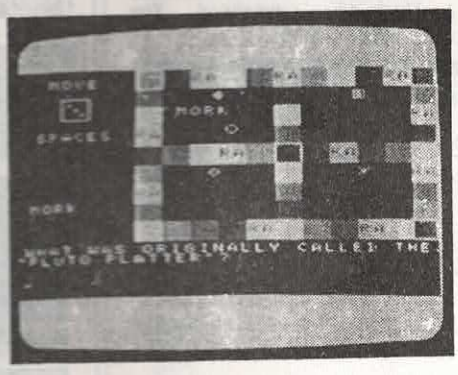
CHOPPER PILOT - The fourteen levels of this game make it very compelling. Keeping clear of canyon walls, fly your helicopter to do battle with enemy fighters, bombers, missiles and UFOs. Great skill required.

UFO (MSX ONLY)

In this machine code game, you are under attack from UFOs. Watch out for the alien unit moving across the bottom of the screen - this will fire 'smart bombs'. Watch out for the meteor shower. Refuel when the truck appears on screen.
Requires 16K RAM, has smooth scrolling, fast response with good sound effects.

METEOR SWARM (SV ONLY)

Another machine code game where your ship is caught in a meteor belt and you must blast your way out. Other ships are caught in the belt and will shoot at you if you do not destroy them first. Comes with bonus game - Birds of Orion.



SPECTRA-TRIVIA (SV 32K)
COMPU-TRIVIA (MSX)

The famous trivia game now on your computer. These are trivia quiz games for up to four players. The computer rolls the dice and each player moves in turn. At certain points you will be asked questions. There is provision for you to create your own question and answer file.

PLEASE SPECIFY SV318/328 OR MSX WITH ORDER.

PEAK_HOUR

As the name suggests, you must try to catch the early morning train, and by the way, watch the fat lady.

This listing is curtesy of :-
------- MSX INTERNATIONAL -------

```
BJ  10 MAXFILES=2
AE  20 SCREEN3,2:COLOR1,1,1:CLS:CLE
    AR1000,&HDD3F:DEFUSR=&HDD40:DEF
    USR1=&HDD50
AD  30 ON STOP GOSUB 1340:STOP ON
GJ  40 RESTORE 70:AD=&HDD40
AB  50 READD$:IF D$<>"*"THEN POKEAD
    ,VAL("&H"+D$):AD=AD+1:GOTO 50 E
    LSE GOTO 60
DN  60 R=RND(-TIME):DEFINT A-Z:DEFF
    NR(A)=INT(RND(1)*A):OPEN"GRP:"A
    S 1
BL  70 DATA 21,80,18,11,00,E1,01,80
    ,01,CD,59,00,C9,00,00,00,06,10,
    11,00,E1,21,01,E1
FC  80 DATA C5,01,1F,00,1A,ED,B0,12
    ,23,13,C1,10,F3,21,00,E1,11,80,
    18,01,80,01,CD,5C,00,C9,*
HM  90 LINE(16,0)-(240,187),10,B:LI
    NE(20,4)-(236,170),7,BF:LINE-(2
    0,183),12,BF
CL  100 LINE(140,140)-(164,170),15,
    BF:LINE(164,140)-(220,170),14,B
    F:LINE(174,150)-(210,160),7,BF:
    LINE(144,150)-(154,170),1,BF:LI
    NE(160,140)-(150,116),6
AN  110 LINE-(210,116),6:LINE-(226,
    140),6:LINE-(136,140),6:PAINT(1
    80,130),6
BK  120 LINE(140,140)-(150,116),9:L
    INE-(160,140),9:LINE-(136,140),
    9:PAINT(150,130),9
DE  130 PSET(32,24),7:PRINT#1,"PEEK
    ":COLOR15:PSET(28,20),7:PRINT#1
    ,"PEEK":COLOR1
AL  140 PSET(32,64),7:PRINT#1,"  HO
    UR":COLOR15:PSET(28,60),7:PRINT
    #1,"  HOUR"
AI  150 GOSUB 640:FOR X=130 TO 0STE
    P-2:C=11:Y=155:GOSUB 1280:FOR W
    =0 TC 200:NEXT:NEXT
AD  160 DEFUSR2=&H41:DEFUSR3=&H44:S
    CREEN2:A=USR2(0):COLOR 15,7,14:
    CLS
BE  170 FOR I=1 TO 30:PLAY"S12M205A
    30":NEXT
CN  180 RESTORE 190:FOR I=1 TO 9:RE
    AD A,B,C,D,E:LINE(A,B)-(C,D),E,
    BF:NEXT
BH  190 DATA 8,60,255,60,1,8,61,255
    ,120,2,8,61,255,64,5,0,121,255,
    191,14,0,0,255,17
BJ  200 DATA 14,0,184,255,191,15,68
    ,70,98,75,15,0,5,255,7,1,0,18,2
    55,20,15
HM  210 FOR I=9 TO 255 STEP 16:LINE
    (I,5)-(I+6,7),15,BF:NEXT
FF  220 PSET(72,69),2:COLOR1:PRINT#
    1,"MSX":LINE(23,0)-(104,12),1,B

BE  230 LINE(24,1)-(103,11),15,BF:P
    SET(38,3),15:PRINT#1,"CENTRAL"
FA  240 FOR I=0 TO 255 STEP16:LINE(
    I,130)-STEP(4,2),15,BF:NEXT
AM  250 FOR B=0 TO 192 STEP 96:LINE
    (B+15,75)-STEP(33,44),1,B
HJ  260 DRAW"BM=B;,59;BR31M-10,-6M+
    10,-6M+10,+6M-10,+6":J=40:CB=2:
    GOSUB 500:GOSUB 510:NEXT
EA  270 FOR I=56 TO 168 STEP 96:LIN
    E(I+1,80)-STEP(46,26),7,BF:LINE
    (I,79)-STEP(48,28),1,B
AB  280 LINE(I,55)-(I+48,59),5,BF:L
    INE(I+1,93)-(I+46,93),1:LINE(I+
    24,80)-STEP(0,26),1:NEXT
AL  290 Q$="C1U8E1U1E3H1U1H1U4E2R1F
    2D4G1D1G1F3D1F1D8"
EI  300 UY=107:FOR J=0 TO 96 STEP 9
    6 :FOR I=59+J TO 100+J STEP 11:
    DRAW"BM=I;,=UY;XQ$;":NEXTI,J
FA  310 GOSUB 1320:LINE(176,0)-(248
    ,12),9,B:LINE(177,1)-(247,11),1
    ,BF:T=29:GOSUB 530
BH  320 A=USR3(0):A=USR(0):FOR I=1
    TO 32:A=USR1(0):FOR W=0 TO 50:N
    EXT:NEXT
AD  330 C$="TOWN HALL CHATSWOOD  HO
    RNSBY   WOY WOY":H=0:N=0:BEEP:P
    UTSPRITE11,(169,3),9,11
DH  340 FOR X=0 TO 116 STEP 4:Y=148
    :C=11:GOSUB 1280:NEXT
DB  350 IF H>4 THEN BEEP:M=4:FOR X=
    X TO 255 STEP 4:GOSUB 1280:NEXT
    :GOTO 1130
HN  360 VDP(1)=&HE2:X=116:Y=148:PX=
    2:PY=120:E=0:CB=2:BEEP:ON INTER
    VAL=360 GOSUB 530
AF  370 '
BG  380 B=FNR(3)*96:INTERVAL ON:IF
    Y<120 THEN E=1
HG  390 IF POINT(X+2,Y+8)=1 ANDPOIN
    T(X+14,Y+2)=1 AND S=1 THEN GOSU
    B 570
HG  400 IF POINT(B+34,99)=2 THEN V=
    FNR(4):FOR J=32 TO 44:CB=1 ELSE
     FOR J=44 TO 32 STEP -1:CB=2
AO  410 GOSUB 500:GOSUB 510:S=STICK
```

```
    (0):ON INT(S/2+.5) GOTO 420,440
    ,450,460
ED  420 IF S=1 AND Y>110 THEN Y=Y-4
    :GOTO 470
JO  430 IF S=1 AND POINT(X+2,Y+8)=1
     ANDPOINT(X+14,Y+2)=1 THEN GOSU
    B 560:GOTO 470
EA  440 IF S=3 AND X<222 THEN X=X+8
    :GOTO 470
DJ  450 IF S=5 AND Y<176 THEN Y=Y+4
    :M=2:GOSUB 1080:GOTO 470
HE  460 IF S=7 AND X>24 THEN X=X-8:
    GOTO 470
EJ  470 C=11:GOSUB 1290:IF PLAY(0)=
    0 THEN PLAY"S4M50001C64"
CH  480 IF E=1 THEN GOSUB 1030
AN  490 NEXT:GOTO 370
AP  500 LINE(B+J,76)-(B+J,118),CB:L
    INE(B+63-J,76)-(B+63-J,118),CB:
    RETURN
AM  510 IF CB=2ANDJ=40 THEN FORI=B+
    19 TO B+40 STEP 21:LINE(I,80)-S
    TEP(4,16),7,BF:NEXT
CO  520 LINE(B+32,76)-(B+32,118),1:
    RETURN
AK  530 T=T+1:LINE(216,1)-STEP(15,1
    0),1,BF:PSET(185,3),1:COLOR 15
DG  540 PRINT#1,USING"AM8:## ";T:IF
     T<60 THEN RETURN ELSE M=3:FOR
    Y=Y TO 174 STEP 4
MD  550 X=X+FNR(20)-10:FOR C=7 TO 1
    0:GOSUB 1290:NEXT:PLAY"S4M50001
    C64":NEXT:GOTO 1080
AP  560 C=9:FOR Y=100 TO 110  STEP5
    :GOSUB 1290:PLAY"S14M20003E64":
    NEXT:MA=ABS(1-MA):RETURN
AH  570 '
BD  580 IF V=0 THEN GOTO 860
FL  590 IF V=1 THEN D=15:F=5:G=6:K=
    6:GOSUB 630:CA=7:PLAY"S0M12000O
    1C1":GOSUB 820
BD  600 IF V=2 THEN D=15:F=9:G=9:K=
    10:GOSUB 630:CA=9:PLAY"S11M1200
    0L1606BFA5":GOSUB 810
FA  610 IF V=3 THEN D=11:F=12:G=5:K
    =13:GOSUB 630:CA=15:PLAY"S0M120
    0001C1":GOSUB 820
CE  620 RETURN
BK  630 PUTSPRITE7,(X-6,90),D,F:PUT
    SPRITE11,(X-6,90),G,K:RETURN
AM  640 RESTORE 660:FOR J=0 TO 13:S
    $="":FOR I=1TO 32:READA$:S$=S$+
    CHR$(VAL("&H"+A$))
CN  650 NEXT:SPRITE$(J)=S$:NEXT:RET
    URN
BC  660 DATA FF,FF,FF,FF,FF,FF,FF,F
    F,FF,FF,FF,FF,00,00,00,00,FF,FF
    ,FF,FF,FF,FF,FF,FF,FF,FF,FF,FF,
    00,00,00,00
AJ  670 DATA 00,00,1F,3F,7F,E3,F3,F
    E,FE,78,3F,C0,40,00,00,00,00,00
    ,F8,FC,FE,C7,CF,7F,7F,1E,FC,03,
    02,00,00,00
AI  680 DATA 00,00,1F,37,77,E3,F3,F
    E,FE,78,3F,C0,40,00,00,00,00,00
    ,F8,EC,EE,C7,CF,7F,7F,1E,FC,03,
    02,00,00,00
CG  690 DATA 3F,7F,E0,C0,80,1C,0C,0
    1,01,07,00,3F,3F,18,70,00,FC,FE
    ,07,03,01,38,30,80,80,E0,00,FC,
    FC,38,1C,1E
BD  700 DATA F5,3F,60,C8,88,1C,1C,0
    1,01,07,00,3F,3F,1C,38,78,57,FC
    ,06,13,11,38,38,80,80,E0,00,FC,
    FC,18,0E,00
GO  710 DATA 0,3,7,5,7,6,7,37,73,E0
    ,80,0,30,78,70,0,0,C0,E0,A0,E0,
    60,E0,EC,CE,7,1,0,C,1E,E,0
BH  720 DATA 7,C,18,18,18,1,0,8,C,1
    F,1F,3F,F,0,0,C0,E0,30,18,18,18
    ,80,0,10,30,F8,F8,FC,F0,0,0,3
FL  730 DATA 7,F,8,16,10,88,9,44,23
    ,1E,7,6,F,8,8,18,C8,F0,20,D0,10
    ,20,20,40,80,FD,C0,C0,E0,20,20,
    30
LO  740 DATA 13,F,4,B,8,4,4,2,1,BF,
    3,3,7,C,18,70,E0,F0,10,68,8,10,
    90,20,C1,7E,E0,60,F0,18,C,7
GE  750 DATA 0,0,0,0,1,1,0,0,0,0,1,
    1,0,0,0,0,0,0,40,E0,D0,70,E0,40
    ,8,8,C0,A0,A0,90,90,0
BK  760 DATA 1,3,F,F,1E,1E,F,16,0,2
    F,1C,0,0,0,0,0,0,E0,F0,B8,18,C,
    C,6,6,E2,F0,38,0,0,0,0,88
IF  770 DATA FE,FE,FE,FE,FE,0,0,0,0
    ,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,
    0,0,0,0,0,0,0,0,0
LF  780 DATA 0,3,F,8,0,8,F,E,7,1,18
    ,20,0,0,0,0,0,E0,F8,8,0,88,F8,3
    8,F2,C4,88,0,0,0,0,0
GK  790 DATA F,1C,10,10,10,0,0,0,0,
    0,7,3,73,77,76,E,F0,1C,4,4,4,0,
    0,0,0,0,70,60,E0,70,30,38
BB  800 '
CP  810 FOR Y=YTO140 STEP 4:FOR W=0
     TO100:NEXT:GOTO 830
FM  820 FORY=Y TO 180 STEP 4
GB  830 X2=X+FNR(20)-10:PUTSPRITE1,
    (X2,Y),CA,2
FJ  840 PUTSPRITE2,(X2,Y),1,4:VDP(1
    )=&HE3:NEXT:FOR W=0 TO 300:NEXT

BB  850 VDP(1)=&HE2:Y=160:PUTSPRITE
    11,(169-H*16,3),9,11:RETURN
AL  860 '
AG  870 INTERVAL OFF:H=H+1:C=11:Y=1
```

```
    00:GOSUB 1280
EF  880 PUTSPRITE2,(X,100),11,4:PLA
    Y"V12L12":FORW=53 TO 78:PLAY"N=
    W;":NEXT
BL  890 Y=93:FOR PX=PX TO 0 STEP-8:
    MA=ABS(1-MA):PUTSPRITE7,(PX-16,
    PY),4,MA+7:FORW=0 TO 50
ED  900 NEXT:NEXT:FORX1=56 TO 87:GO
    SUB1010:NEXT:FOR X1=164 TO 182:
    GOSUB 1010:NEXT
DM  910 FORX1=182 TO 154 STEP -1:GO
    SUB 1010:NEXT
CJ  920 FOR X1=87 TO 64 STEP -1:GOS
    UB 1010:NEXT
FP  930 IF PLAY(0) THEN 930
AC  940 GOSUB 1110
FD  950 GOSUB 1320:FORI=1 TO 32:X1=
    64-I*8:A=USR1(0):GOSUB 1010:FOR
     W=0 TO 90:NEXT:NEXT
CL  960 LINE(28,1)-(102,11),15,BF:P
    SET(28,3),15:COLOR1:PRINT#1,MID
    $(C$,H*10-9,10)
DK  970 BEEP:IF H<5 THENPUTSPRITE11
    ,(169-H*16,3),9,11
AD  980 FOR B=0 TO 192 STEP96:FORJ=
    32 TO 44:CB=1:GOSUB 500:NEXT:GO
    SUB 510:NEXT
KC  990 PLAY"T255V13O6L64BAGFEDCL16
    03":PUTSPRITE0,(0,209):FORY=YTO
    148 STEP2
CP  1000 C=11:GOSUB 1290:NEXT:GOTO
    350
DC  1010 MA=ABS(1-MA):PUTSPRITE1,(X
    1,Y),11,MA+1:PUTSPRITE2,(X1,Y),
    1,MA+3
FJ  1020 PUTSPRITE0,(X1+MA,Y+14),2,
    0:RETURN
BN  1030 MX=(SGN(X-PX)*2):MY=(SGN(Y
    -PY)*2):PX=PX+(MX*(PX>0)*(PX<25
    5))
DJ  1040 PY=PY+(MY*(PY>100)*(PY<180
    )):PUTSPRITE7,(PX,PY),4,MA+7
FF  1050 IF(PX-10<X)*(PX+10>X)*(PY-
    8<Y)*(PY+12>Y) THEN GOSUB 1070
BH  1060 RETURN
BM  1070 BEEP:Y=Y+8:C=11:GOSUB 1280
    :M=1
CH  1080 IF Y<168 THEN RETURN ELSE
    Y=184:C=15:GOSUB 1290
EN  1090 PLAY"S0M500002A":FORJ=1 TO
     20:FORI=1 TO 14:COLOR 15,4,I:N
    EXTI,J:GOTO 1130
CH  1100 '
CM  1110 FOR B=0 TO 192 STEP96:FORJ
    =44 TO 32 STEP-1:CB=2:GOSUB 500
    :GOSUB 510:NEXT:NEXT
EH  1120 FORI=0 TO 10:PUTSPRITEI,(0
    ,209):NEXT:RETURN

AI  1130 INTERVAL OFF:PUTSPRITE11,(
    0,209):LINE(0,0)-(255,24),7,BF:
    GOSUB 1120:GOSUB 1110
CL  1140 ON M GOTO 1150,1160,1170,1
    180
LC  1150 M$="  PLEASE CATCH A TAXI
    ":GOSUB 1190
KD  1160 M$="  PLEASE CATCH A BUS
    ":GOSUB 1190
GO  1170 M$="  WALK IT'S HEALTHIER
    ":GOSUB 1190
II  1180 M$="  YOU ARE AT GOSFORD !
    ! "
JI  1190 GOSUB1320:FORI=1TO 32:LINE
    (I*8-8,0)-(I*8,120),7,BF:GOSUB
    1300:A=USR1(0):NEXT
BJ  1200 BEEP:COLOR1:IF M<>4 THEN L
    INE(38,90)-(210,98),1,BF:PSET(7
    0,60),7
GG  1210 IF M<>4 THEN PRINT#1,USING
    "START 9:10  TIME=9:## ";T-30:P
    SET(40,90),1:COLOR15:PRINT#1,M$

AP  1220 IF M=4 THEN PSET(70,60),7:
    PRINT#1,USING"START 9:10  TIME
    AM8:## ";T
HE  1230 PSET(90,140),14:COLOR1:PRI
    NT#1,"REPLAY(Y/N)?"
EA  1240 IF INKEY$<>""THEN 1240
GC  1250 IF INKEY$="Y" OR INKEY$="y
    " THEN 160
DB  1260 IF INKEY$="N" OR INKEY$="n
    " THEN CLS:GOTO 1340
AH  1270 GOTO 1250
FP  1280 PLAY"S4M50001C64"
GL  1290 MA=ABS(1-MA):PUTSPRITE1,(X
    ,Y),C,MA+1:PUTSPRITE2,(X,Y),1,M
    A+3:RETURN
HC  1300 IF M=4 AND I>8THEN LINE(32
    +(I-8)*8,90)-STEP(8,8),4,BF ELS
    E RETURN
HE  1310 PSET(32+(I-8)*8,90),4:COLO
    R9:PRINT#1,MID$(M$,I-8,1):RETUR
    N
FI  1320 RESTORE 1330:FORI=0 TO 13:
    READD:SOUNDI,D:NEXT:RETURN
FP  1330 DATA 111,1,110,1,109,1,10,
    56,15,15,15,0,0,11
GL  1340 SCREEN0:KEYON:END
```

THE END

CHRISTMAS

By Hank Zuydam.
This programm won the Music and Graphics section of our Competition

```
GG  10 COLOR15,4,0:SCREEN1,3
EJ  20 ONSTOP GOSUB270:STOPON
AF  30 CO=2:GOSUB280:GOSUB210
EC  40 FORX=1TO2:LOCATE30+X,15:PRIN
    T"MERRY"
BH  50 LOCATE5+X,35:PRINT"CHRISTMAS
    ":NEXT
ED  60 FORX=1TO15:GOSUB190:Y=Y+2:NE
    XT
BD  70 FORX=10TO30:GOSUB190:Y=Y+2:N
    EXT
AP  80 FORX=20TO45:GOSUB190:Y=Y+2:N
    EXT
GL  90 FORX=40TO60:GOSUB190:Y=Y+2:N
    EXT:CO=7:X=3
KM  100 FORY=YTO172:GOSUB190:Y=Y+1:
    NEXT:CO=9
BD  110 FORX=15TO10STEP-.5:GOSUB190
    :Y=Y+2:NEXT
CH  120 PRINT:PRINT"      FROM "
BM  130 PRINT:PRINT" HANK & APRIL"
DI  140 PRINT:PRINT"      TO"
EN  150 PRINT:PRINT"  EVERYBODY"
AN  160 PLAYA$,B$,C$:GOSUB360:PLAYA
    $,B$,C$:GOSUB360
AJ  170 PLAYD$,F$:GOSUB360:
AB  180 GOTO160
MP  190 LINE(128-X,Y  )-(128+X,Y  )
    ,CO,BF
BF  200 LINE(128-X,Y+1)-(128+X,Y+1)
    ,CO+1,BF:RETURN
CP  210 A$="T255L3o5Co4L8B-L4AGFGAF
    L8GAB-GL3AL8GL4FEL2F"
AP  220 B$="T255L3o4AL8GL4FEdEFCL8E
    FGEL3FL8DL4CCL2C"
AG  230 C$="T255L3o3FL8EL4FCDCFFL8C
    CCCL3FL8o2B-L4o3CCL2F"
HA  240 D$="T255L3o4Go4L8AL4B-GL3AL
    8B-o5L4Co4GL8AB-o5L4CL8DEL4FEDL
    2C"
NM  250 F$="T255L3o3CL8CL4CCL3FL8FL
    4FEL8CDL4CL8o3B-o4CL4CCo3B-L2G"

BM  260 RETURN
EN  270 COLOR15,4,5:SCREEN0,1:RUN
CB  280 FORA=1TO32:READVV$:V$=V$+CH
    R$(VAL("&H"+VV$)):NEXTA
AG  290 SPRITE$(28)=V$
BI  300 FORA=1TO32:READWW$:W$=W$+CH
    R$(VAL("&H"+WW$)):NEXTA
AD  310 SPRITE$(29)=W$

BK  320 FORA=1TO32:READXX$:X$=X$+CH
    R$(VAL("&H"+XX$)):NEXTA
AK  330 SPRITE$(30)=X$
BM  340 FORA=1TO32:READYY$:Y$=Y$+CH
    R$(VAL("&H"+YY$)):NEXTA
DG  350 SPRITE$(31)=Y$:RETURN
CD  360 IFA=0THEN390
CA  370 A=0:FORX=1TO80:GOSUB410:NEX
    T
CB  380 A=0:RETURN
EC  390 FORX=80TO1STEP-1:GOSUB410:N
    EXT
CJ  400 A=1:RETURN
GB  410 PUTSPRITE28,(220-X,127),6,2
    8
DE  420 PUTSPRITE29,(220-X,127),15,
    29
FO  430 PUTSPRITE30,(220-X,159),6,3
    0
ED  440 PUTSPRITE31,(220-X,159),15,
    31
BP  450 RETURN
CA  460 DATA00,C0,E1,03,E0,E3,E0,E1
    ,E0,E0,FC,FC,7E,3E,0F,0F
BP  470 DATA38,F8,F8,F8,00,F8,00,20
    ,80,80,0C,0E,1F,1F,37,F7
CO  480 REM
AC  490 DATA00,00,00,E0,03,00,03,02
    ,03,03,03,03,01,01,00,00
DA  500 DATA06,06,00,00,F8,00,F8,D8
    ,78,78,F0,F0,E0,E0,C0,00
CG  510 REM
AP  520 DATA1F,1F,3F,3F,00,00,1E,1E
    ,1E,1E,1E,1E,00,00,00,00
AC  530 DATAFF,FF,FF,F8,07,03,F0,F0
    ,F0,F0,F0,F0,00,00,00,00
BB  540 DATA00,00,00,00,7F,7F,00,00
    ,00,00,00,00,1C,1C,7D,7D
BJ  550 DATA00,00,00,07,F8,F8,00,00
    ,00,00,00,00,70,70,F0,F0
AN  560 CLS:RESTORE:GOTO10
```

THE END

ALLIGATOR

By Peter White.

This is the programm that won the Music section of the competition.
¼ = [LEFT GRPH]+M (ASCII 172)

```
GK  10 COLOR 15,1,1:SCREEN 1:COLOR
    15
HL  20 LOCATE83,70:PRINT"Alligator
    Crawl"
LO  30 LOCATE 83,78:PRINT"¼¼¼¼¼¼¼¼
    ¼¼¼¼¼¼"
BK  40 COLOR 10:LOCATE 29,88:PRINT
    "Original by ";:COLOR 15:PRINT
    "Thomas 'Fats' Waller."
DK  50 COLOR 10:LOCATE 41,98:PRINT
    "Computer version arranged and"

AC  60 LOCATE87,108:PRINT"programme
    d by:"
BM  70 COLOR 15:LOCATE 95,118:PRINT
     "Peter White"
EK  80 REM¼¼¼¼¼¼¼¼¼¼¼¼¼¼¼¼¼¼¼¼¼¼¼
    ¼¼¼¼¼¼
EJ  90 REM¼¼¼¼¼¼¼¼¼¼¼¼¼¼¼¼¼¼¼¼¼¼¼
    ¼¼¼¼¼¼
NL  100 A$="l16o1c8.o2co1e8.o2eo1g8
    .o2go1a8.o2ac8.o3co1a8.o2ao1g8.
    o2go1e8.o2e"
FK  110 B$(1)="o2l16c8.o3co2e8.o3eo
    2g8.o3go2a8.o3ac8.o4co2a8.o3ao2
    g8.o3go2e8.o3e"
AL  120 B$(2)="o5l12e-dco4a8"
IN  130 B$(3)="o5g12"
DG  140 C$(1)="o2l4e-o3g-o1a-o3g-"
CG  150 C$(2)="o5rl12g-ff-e-8g-4o6e
    -8"
FF  160 C$(3)="o6re-4c8e-4o5g-8"
CJ  170 C$(4)="o5l12g-ff-e-8g-4.r"
AL  180 C$(5)="o6e-4c8e-4.r"
FP  190 D$(1)="o2l4go3go2ao3a"
ED  200 D$(2)="o2l4do3f#o1go3f"
NN  210 D$(3)="o6l16r8.d#e8.o5bo6d8
    .co5bo6co5ba"
DM  220 D$(4)="o5b8.o6c16d8g4.r"
HG  230 D$(5)="o5b8.o6c16d8o5g4.r"
EE  240 E$(1)="o3c1c8r8o2g8c4.r4"
JD  250 E$(2)="o3l8r8.e16f.g16a.b16
    o4c.d16er8fg4.r4"
AP  260 E$(3)="o4l8r8.c16d.e16f.g16
    a.b16o5cr8de4.r4"
ME  270 PLAY "t150v9","t150","t150"
JP  280 PLAY A$+A$:PLAY "v8"
CC  290 PLAY B$(1),B$(2),B$(3)
BL  300 PLAY C$(1),C$(2),C$(3)
BI  310 PLAY C$(1),C$(4),C$(5)
AN  320 PLAY D$(1),D$(3)
AL  330 PLAY D$(2),D$(5)
AP  340 PLAY D$(1),D$(3)
AO  350 PLAY D$(2),D$(4)
BO  360 PLAY B$(1),B$(2),B$(3)
CC  370 PLAY C$(1),C$(2),C$(3)
BP  380 PLAY C$(1),C$(4),C$(5)
BE  390 PLAY D$(1),D$(3)
AH  400 PLAY D$(2),D$(5)
CB  410 PLAY E$(1),E$(2),E$(3)
IE  420 REM¼¼¼¼¼¼¼¼¼¼¼¼¼¼¼¼¼¼¼¼¼
    ¼¼¼¼¼¼¼
ID  430 REM¼¼¼¼¼¼¼¼¼¼¼¼¼¼¼¼¼¼¼¼¼
    ¼¼¼¼¼¼¼
GF  440 A$(1)="o3l4cgo2b-o3go2ao3f#
    o2a-o3f"
CI  450 A$(2)="o5l16c8.dd#8.ec8.dd#
    8.ef#4o6e8d8d2"
BO  460 A$(3)="o5l16e8.ff#8.ge8.ff#
    8.go6e4o5f#8f8f2"
BA  470 A$(4)=D$(1):A$(5)=D$(3)
FM  480 B$(1)="o6d4o5b8g8g2"
GG  490 B$(2)="o2l4do4fo2go4f"
HD  500 B$(3)="o2l4do3ao2go3g"
AH  510 C$(1)="o2l4go3go1go3gco2g8c
    4."
NL  520 C$(2)="o4l4rerfe2."
GN  530 C$(3)="o6l16c8o5g8egeca8f8d
    fdo4bo5c2."
DL  540 D$(1)="o2l4eo3go1bo3go2eo3g
    o1bo3go2eo3go2do3go2c#o3go1ao3g
    "
OH  550 D$(2)="o5l16r8.d#e8.o4bo5d#
    8e4d#8e8.o4bo5d#8e8e8.d#e8.o4bo
    5d#8e4d#8e4o4b4g8.o5c#e8.gb8.ge
    8.c#"
KG  560 D$(3)="o5l16r8.a#b8.r16a#8b
    4a#8b8.r16a#8b8b8.a#b8.r16l4a#8
    ba#8bge8.r16o4ere"
BC  570 E$(1)="o2l4do3ao1ao3ao2do3a
    o1ao3ao2do3ao2ao3al16g8r16go2g8
    r16go3g8r16go2g8r16g"
FG  580 E$(2)="o5l8r8.c#16d.o4a16o5
    c#d4c#d.o4a16o5c#dd.c#16d.o4a16
    o5c#d4c#d4o4g4f2o5g2"
DI  590 E$(3)="o5l8r8.g#16a.r16g#a4
    g#a.r16g#aa.g#16a.r16g#a4g#a4c#
    4d2o4f2"
GL  600 F$(1)="o2l4ao3ao2g#o3g#"
AH  610 F$(2)="o5c2o4b2"
DO  620 F$(3)="o5l12fee-d4fee-d4"
GM  630 F$(4)="o2l4go3go2co4c"
AE  640 F$(5)="o4b-2r8o5a4."
DM  650 F$(6)="o5l12fee-d8a4e-4."
GJ  660 F$(7)="o2l4go3go4co3c"
FM  670 F$(8)="o4b-2e2"
AG  680 F$(9)="o5l12fee-d4c2"
JI  690 G$(1)="o2l12gb-o3d-fgb-o4d-
    4c4o3l16f8.ed8.ce8.dc8.o2b-"
```

```
EE  700 G$(2)="o3l12gb-o4d-fgb-o5f4
    e8c8c2"
EB  710 G$(3)="o4r2b-4g8f8f2"
LL  720 H$(1)="o2l16g8.ab-8.bo3c4o2
    c4f2o3g2"
KC  730 H$(2)="o6l16f8.dr4o5f4e4f2d
    #2"
AP  740 H$(3)="o6l16d8.o5b-d8.fa4o4
    b-4a2b2"
BG  750 PLAY A$(1),A$(2),A$(3)
AC  760 PLAY A$(4),A$(5)
BL  770 PLAY B$(1),B$(2),B$(3)
BJ  780 PLAY A$(1),A$(2),A$(3)
CA  790 PLAY C$(1),C$(2),C$(3)
CB  800 PLAY "o1b4","o2b4"
BK  810 PLAY D$(1),D$(2),D$(3)
BO  820 PLAY E$(1),E$(2),E$(3)
BD  830 PLAY A$(1),A$(2),A$(3)
AB  840 PLAY A$(4),A$(5)
BI  850 PLAY B$(1),B$(2),B$(3)
BG  860 PLAY A$(1),A$(2),A$(3)
BN  870 PLAY C$(1),C$(2),C$(3)
CG  880 PLAY "r","r","r"
CI  890 PLAY F$(1),F$(2),F$(3)
BF  900 PLAY F$(4),F$(5),F$(6)
BP  910 PLAY F$(1),F$(2),F$(3)
AO  920 PLAY F$(7),F$(8),F$(9)
CB  930 PLAY F$(1),F$(2),F$(3)
BJ  940 PLAY F$(4),F$(5),F$(6)
CG  950 PLAY G$(1),G$(2),G$(3)
CE  960 PLAY F$(1),F$(2),F$(3)
BM  970 PLAY F$(4),F$(5),F$(6)
CG  980 PLAY F$(1),F$(2),F$(3)
BF  990 PLAY F$(7),F$(8),F$(9)
FI  1000 PLAY F$(1),F$(2),F$(3)
FA  1010 PLAY F$(4),F$(5),F$(6)
GA  1020 PLAY H$(1),H$(2),H$(3)
EM  1030 PLAY A$(1),A$(2),A$(3)
DI  1040 PLAY A$(4),A$(5)
FB  1050 PLAY B$(1),B$(2),B$(3)
EP  1060 PLAY A$(1),A$(2),A$(3)
FG  1070 PLAY C$(1),C$(2),C$(3)
CA  1080 PLAY "o1b4","o2b4"
FL  1090 PLAY D$(1),D$(2),D$(3)
FE  1100 PLAY E$(1),E$(2),E$(3)
EJ  1110 PLAY A$(1),A$(2),A$(3)
DF  1120 PLAY A$(4),A$(5)
EO  1130 PLAY B$(1),B$(2),B$(3)
EM  1140 PLAY A$(1),A$(2),A$(3)
FD  1150 PLAY C$(1),C$(2),C$(3)
EM  1160 PLAY "r4","r4","r4"
AM  1170 IF PLAY(1) THEN 1170
```

THE END

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